Can activation-derived feedback improve prompt optimization?

Current pilot design | Extra descriptions enter training reflection only

A. Collect paired answer and pre-answer activations on training questions

Training question + current prompt

Source model Gemma 3 27B

Actual answer

Gold / reference (judge only)

GPT Well judge FPQ / NFP criteria

Shared text feedback Question, answer, score, judge rationale

States are captured before the first answer token.

Same generation's pre-answer states Question-end positions

Fixed NLA or AO reader Activation input only No question / gold text

Extra train feedback NLA / AO description Corresponding arm only

NLA: decoder block 41, one question-end vector | AO: block 31, last up to 20 question-token vectors

B. Compare four GEPA arms with the same seed prompt, data and search limits

GEPA
Shared feedback

+ Text diagnostic
From shared text feedback

+ NLA description
From activations

+ AO description
From activations

GPT reflection → Candidate prompt → Fresh source answer + states → Train feedback ↺

C. Select on GPT dev scores, freeze, then evaluate independently

GPT candidate-dev scores Select best prompt

Freeze prompt No Haiku reselection

Held-out answers → Haiku Source uses selected prompt only

FPQ correction + NFP preservation Report both separately

Pilot: 100 train questions across two datasets; 8 candidate-dev + 8 separate held-out questions per dataset. Held-out pilot ≠ official test.